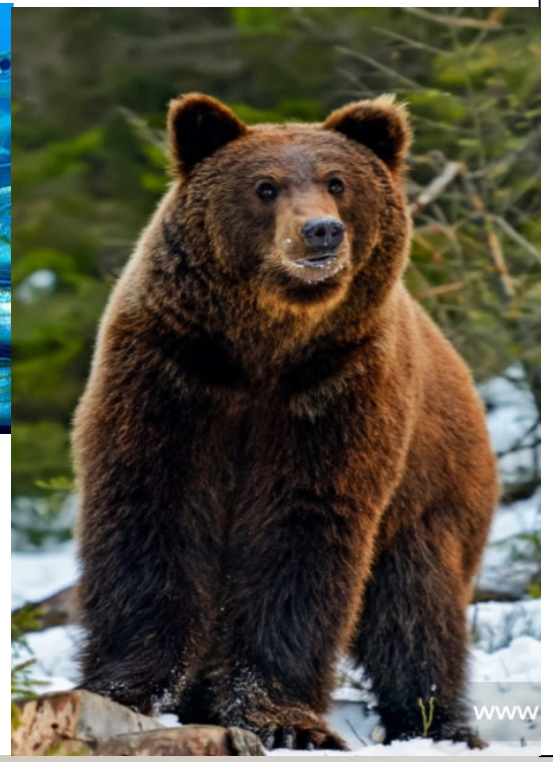
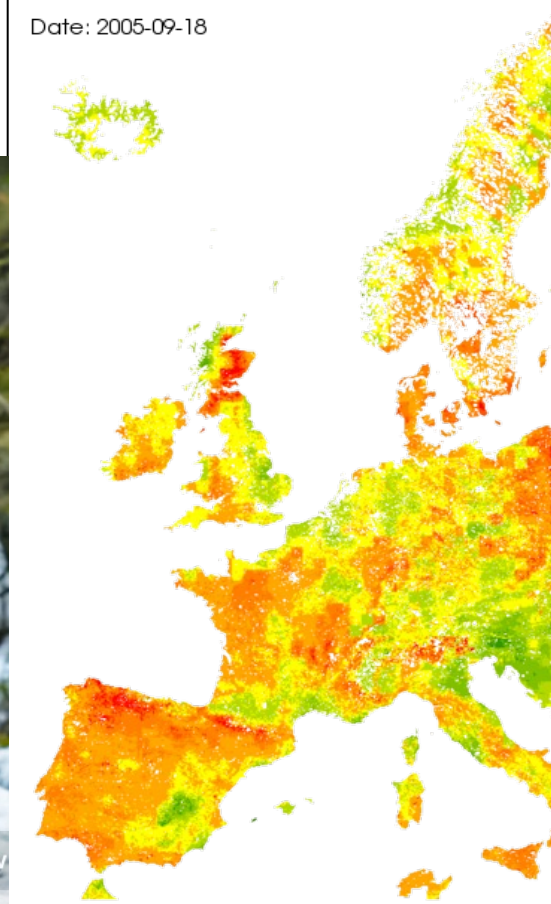
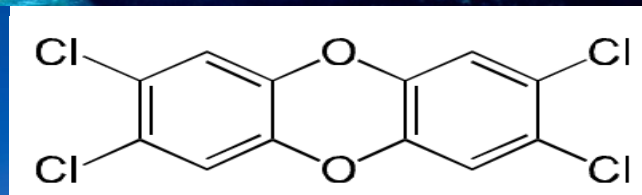
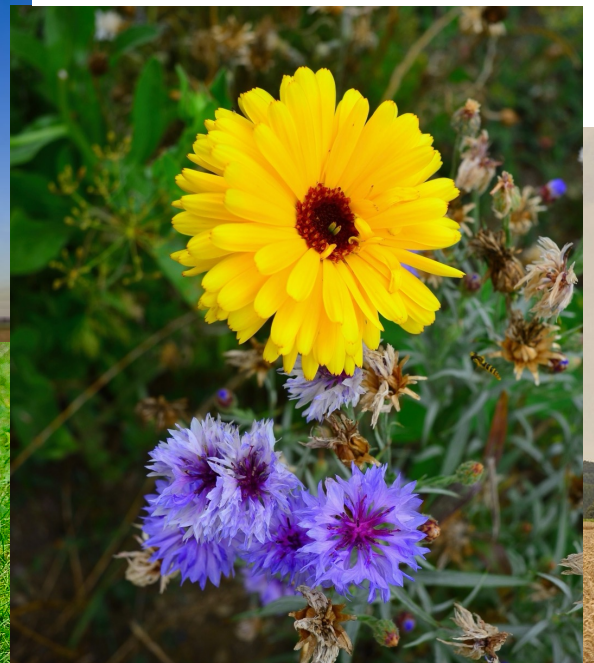
Ch. 5.2: The Solidarity Policies.









Ch. 5.2: The Solidarity Policies.

Vincent van Gogh: Landscape near Auvers 1880 (Van Gogh Museum Amsterdam)

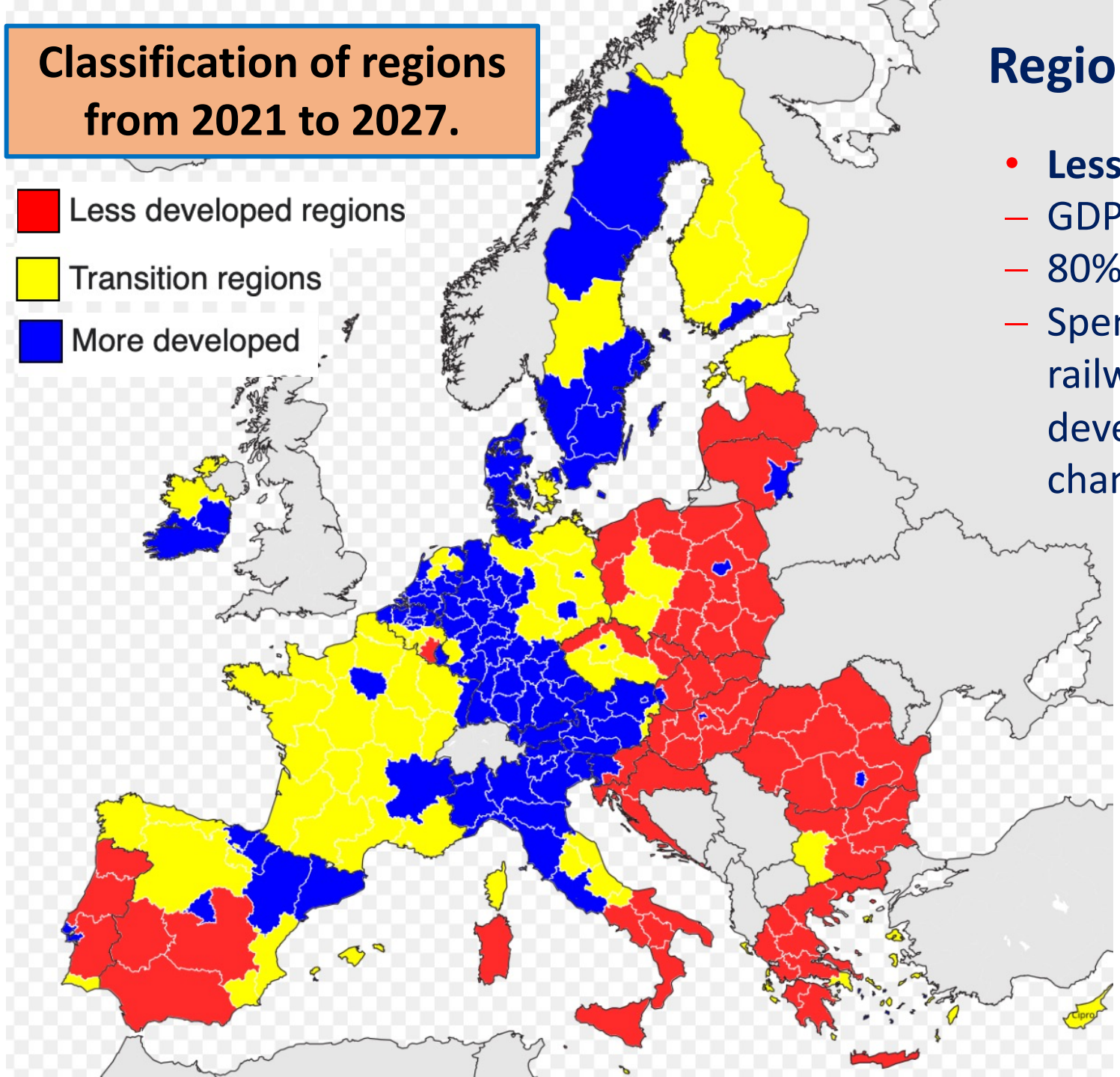
Regional Aid and Cohesion Policies.

- **Purpose of solidarity policies**
  - support the completion of the single market,
  - correct structural and economic imbalances,
  - support industrial sectors encountering difficulties.
- **Regional aid**
  - transfers of funds from rich to poor countries/regions:
    30 billion EUR annually (25 % of the EU budget)
  - main 'instruments' (like investment tools) are:

- 'European Regional Development Fund', which has provided poorer regions with billions of Euro to improve job prospects and help local businesses.
- 'Cohesion Fund' is used to finance transport and environmental initiatives in Europe's less well-off regions.
- European Investment Bank provides about € 100 billion ***in loans*** annually for projects to improve transport and telecommunications infrastructures, to provide industrial development initiatives, to develop urban areas and to support health and education.

Regional Aid and Cohesion Policies.

Classification of regions from 2021 to 2027.



- **Less developed regions:**
  - GDP per capita under 75% of the EU average.
  - 80% of the funds are spent on this objective.
  - Spending mostly on infrastructure: roads, railways, scientific institutions, cultural development of cities, adaptation to climate change.

20% of the Funds for Regional Development and Cohesion Policies are given to the transition regions and more developed regions for special projects which are in the Community interest.

Use of EU Solidarity Funds.

Regional aid projects 2000- 2006:

- **Spain:** 35 billion EUR
- 1.200 km new roads and motorways
- 850 km of high speed rail network
- 600 km new water pipelines
- 4 billion EUR investment in R&D and innovation
- **Italy:** 27 billion EUR
- 20.000 companies in the South supported
- 563 cultural heritage projects in the South
- 350 km railways modernised
- **Austria:** 800 million EUR
- 900 R&D projects
- 1.150 environmental projects

Urban renovation in Vienna supported by the Communiy Initiative on Urban Areas

The Common Agricultural Policy (CAP).

- The CAP provides subsidies for the European farmers and consumes ca. one third of the budget of the European Union.
- This is subject to severe criticism whenever a new EU budget is set up, since the contribution of agriculture to the GDP (Gross Domestic Product = Economic Output) is only 1,5% (industry 20%).
- European food and feed production is costly due to the small structure of European farms (factor 2 compared to the USA).
- In addition increasing pressure on EU farmers due to globalization due to liberalization of

•Original aims of the CAP (1957):

-to achieve a fair standard of living for the farming community,

-to stabilize the markets,

-to supply the consumers at reasonable prices, and

-to modernize farming infrastructure.

The Common Agricultural Policy (CAP).

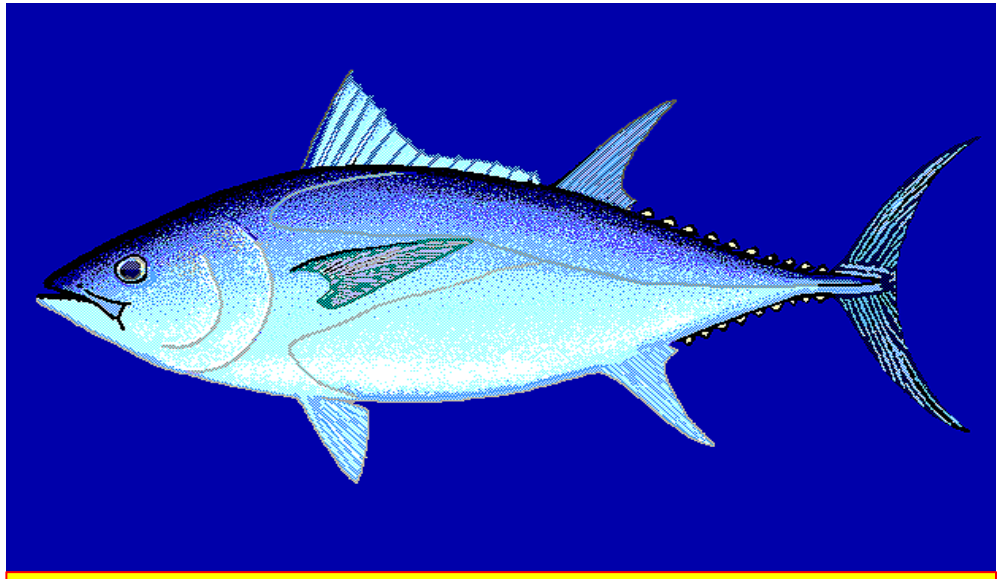
- These objectives have been fully achieved.
- Europe produces more food than it can consume while in the Sixties it had to import substantial amounts of grain.
- Agricultural productivity has dramatically gone up: While in the Sixties 40 % of the population worked in farming it is now only 3 %.
- Consumers spend only about 10 % of their income for food now compared to nearly 50 % in the Sixties.

- Financial support now largely decoupled from the production volume and environmental criteria have been introduced:
- Sustainable development of rural areas and protection of the diversity of Europe's countryside.
- Use of sustainable farming practices.

The Common Fisheries Policy.

- The European Union Common Fisheries Policy (CFP), has been established because 80 % of the fish stocks in EU waters are overfished.
- **Aim of the CFP:**
  - reduce the overcapacity in fishing fleets,
  - preserve fish stocks,
  - provide financial assistance to allow fishing communities to develop other economic activities.
- **Measures of the CFP:**
  - assignment of lots for fishing operations,
  - limitation of number of fishing vessels,
  - quota for catches.

Bluefin tuna, the most valuable species: 60.000 tons caught annually in the Mediterranean and Atlantic, but the scientifically recommended catch is only 15.000 tons maximum.



Bluefin tuna value 400.000$.

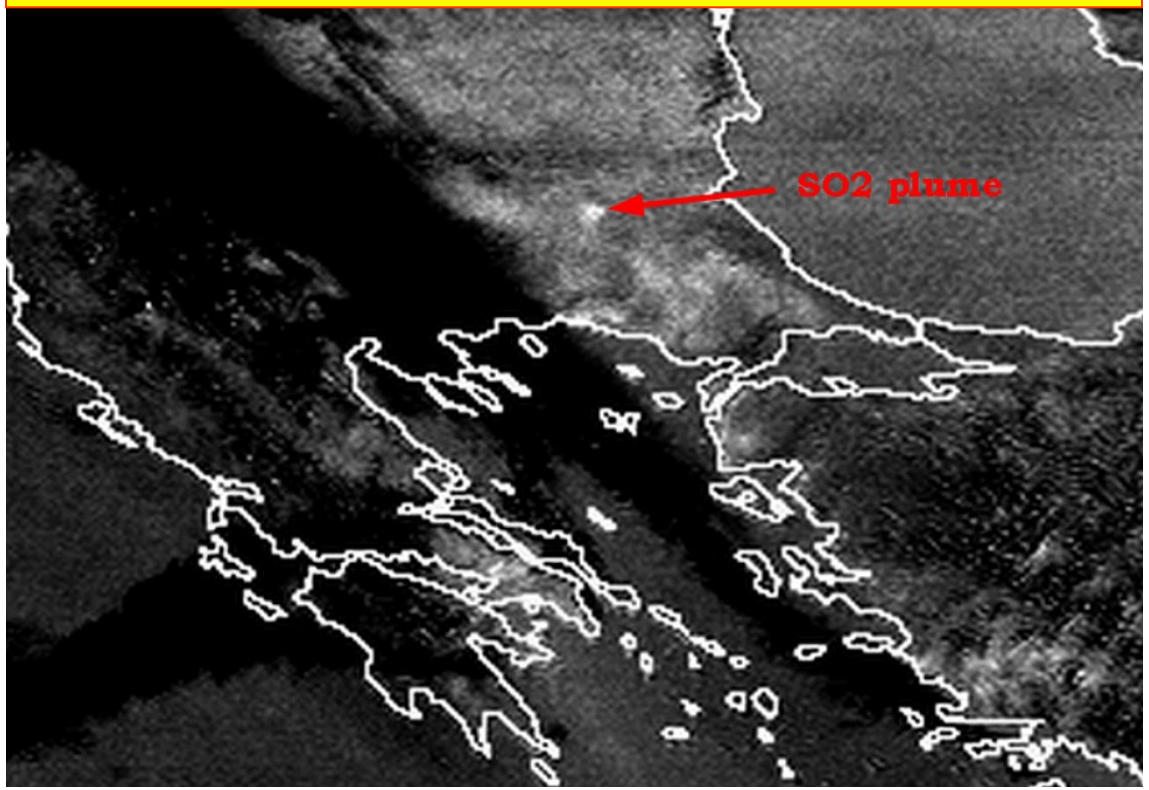
Environmental Policy.

- Environmental policy of the European Community since 1972.
- Environmental protection became a Community Matter with the Treaty of Amsterdam.
- 8 Environmental Action Programs with more than 300 pieces of important environmental legislation address all pertinent issues such as acid rain, the thinning of the ozone layer, air quality, noise pollution, waste management, water pollution, biodiversity or climate change.

- EU Policies provided a high environmental quality.
- Acquis contains strict environmental regulations leading to a great improvement of the situation in new Member States.
- The overall policy concept of the EU, Sustainable Development, has become a model for the whole world.

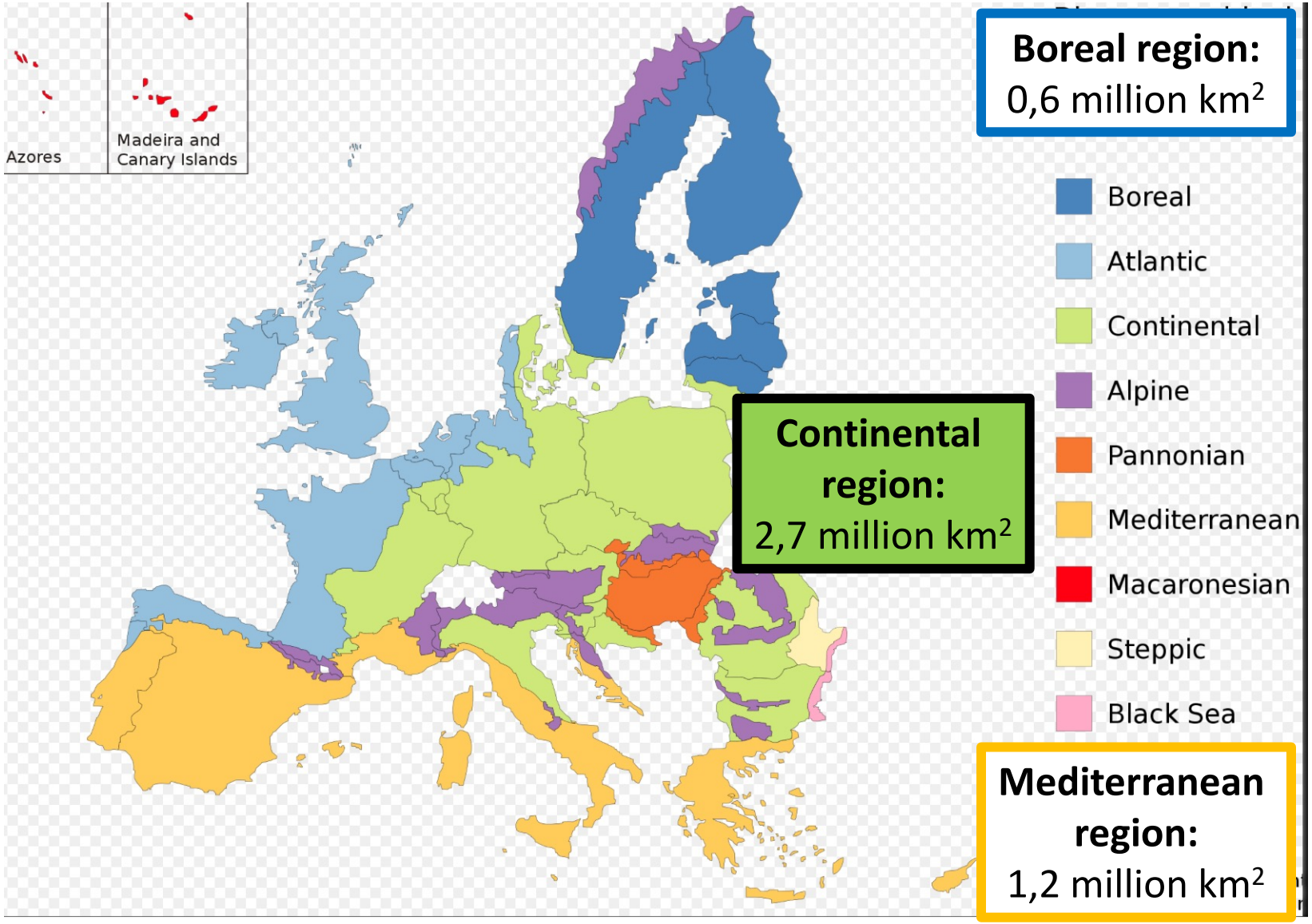
Coal fired power plants emitted ca 2.200 tons SO_2 per day. Source: Meteosat 2005

SO_2 emissions in Stara Zagora (Bulgaria)



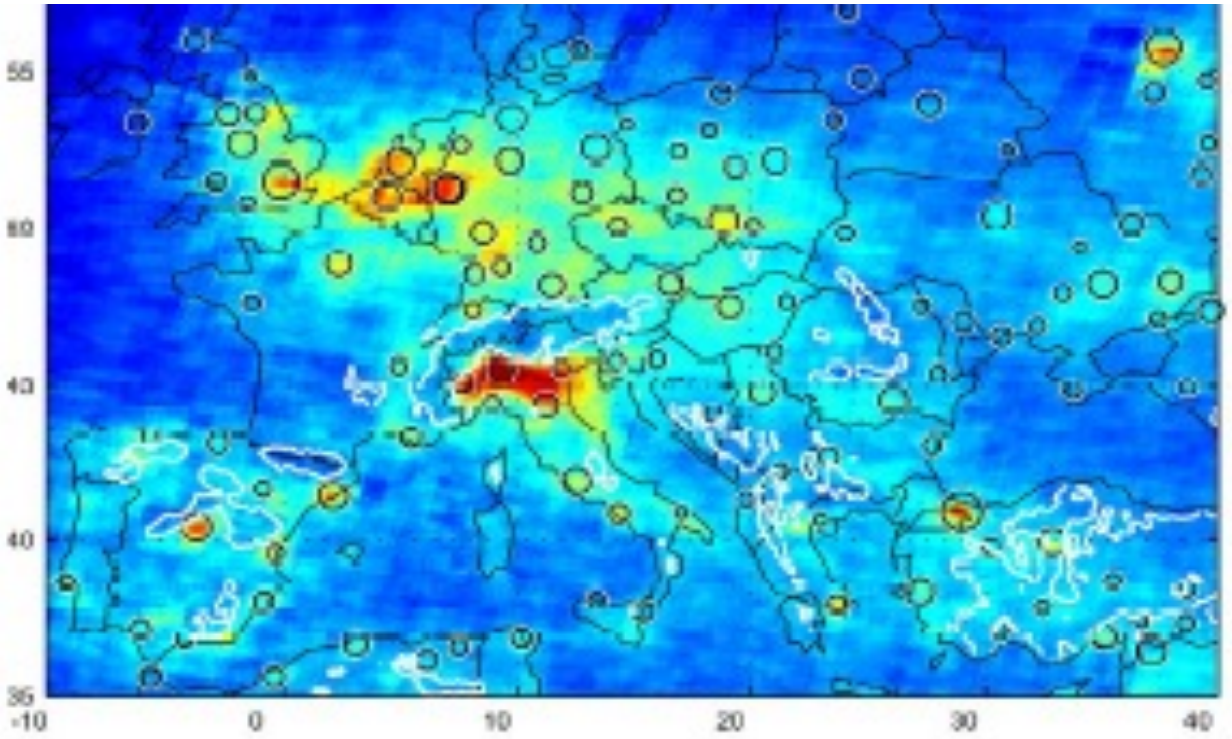
A Differentiated Environmental Policy.

Biogeographical regions of Europe:
One Environmental Policy from the Arctic to the Mediterranean.



Clean Air for Europe.

- Industrial air pollution has been dramatically reduced (NEC Directive).
- Road traffic is the now major source for air pollution.
- The situation has however also substantially improved over the last decades.



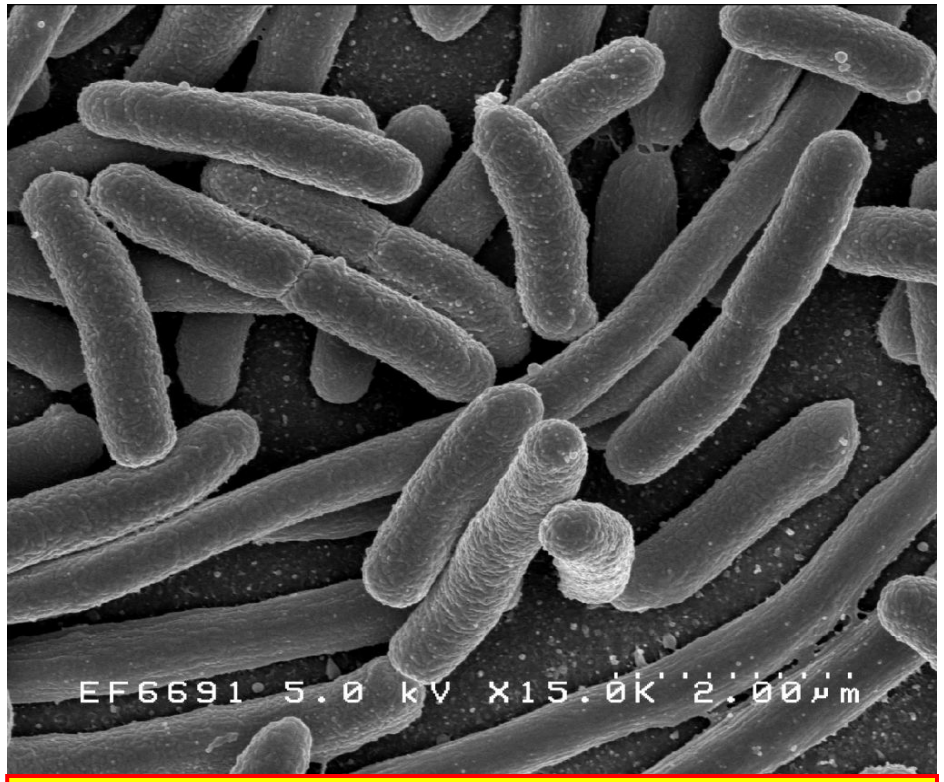
Tropospheric NO_2. Source Beirle 2004

- Community measures:
  - Lead and sulfur free gasoline and diesel.
  - Catalytic converters for gasoline cars.
  - Particle filters for diesel passenger cars.
  - Maximum emission limits for passenger cars and large vehicles.
  - Maximum fuel consumption for the fleet of a passenger car manufacturer of 4,5 l/100km.

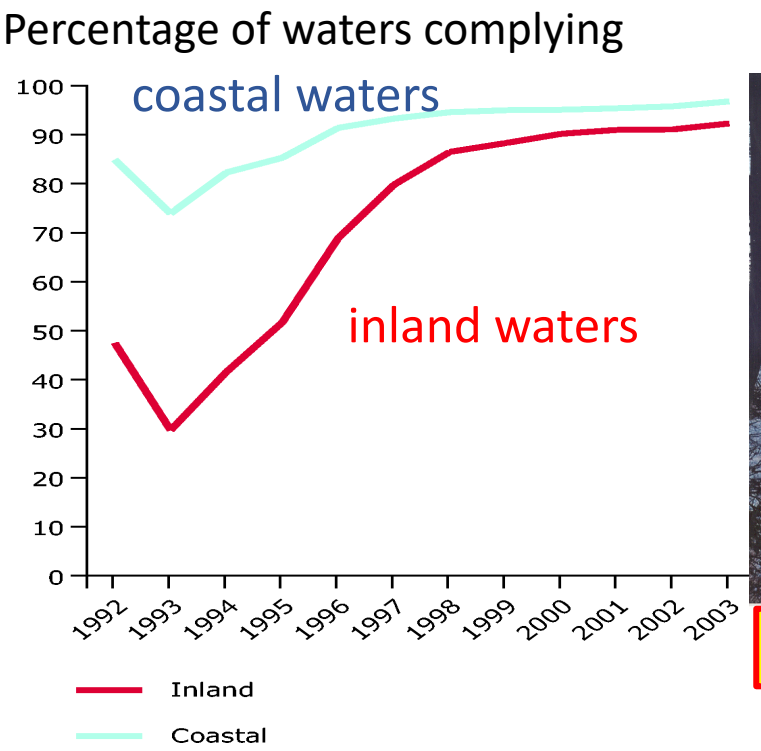
- The emission reduction per vehicle over the last 20 years amounts to 95%.

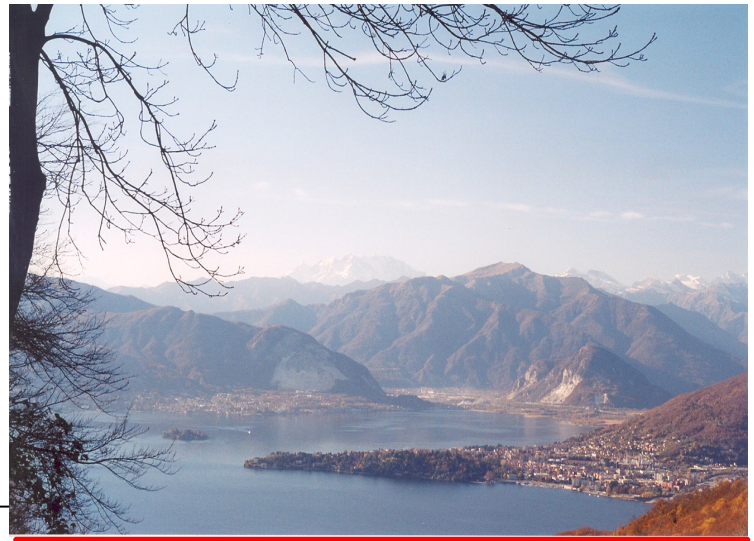
Quality of Bathing Waters in the EU.

- In the 1980s the quality of the water at Europe's beaches and inland waters was a serious concern.
- Main problem: bacterial contamination and algae growth due to excessive input of nutrients (from agriculture and households).
- Directives 76/160/EC and 2006/7/EC ensure a steady improvement of the quality of inland and coastal bathing waters.



Escherichia coli cells magnified 25,000x.





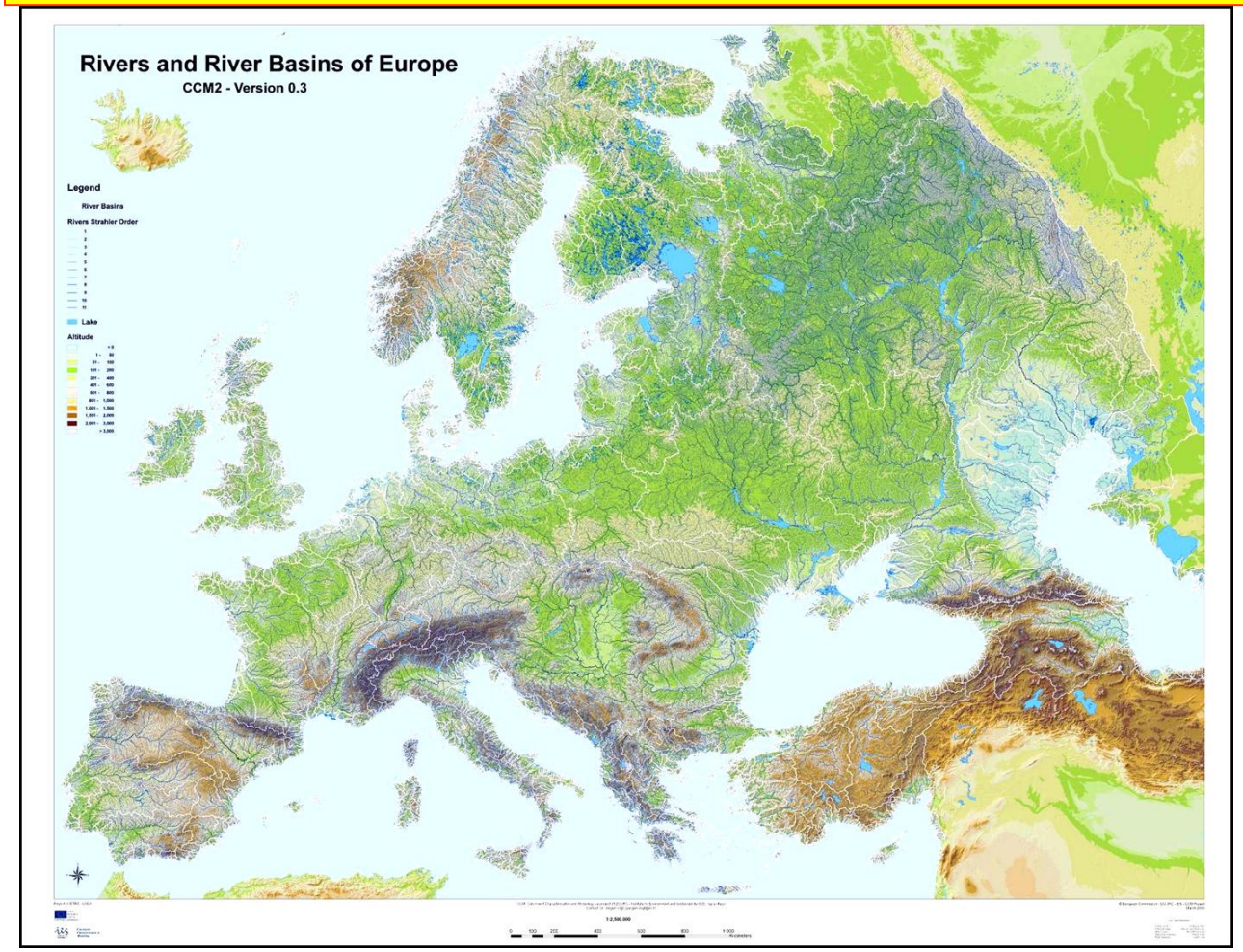
Lago Maggiore.

- Already in 2003, 97% of coastal bathing waters and 92% of inland bathing waters complied with the mandatory standards.

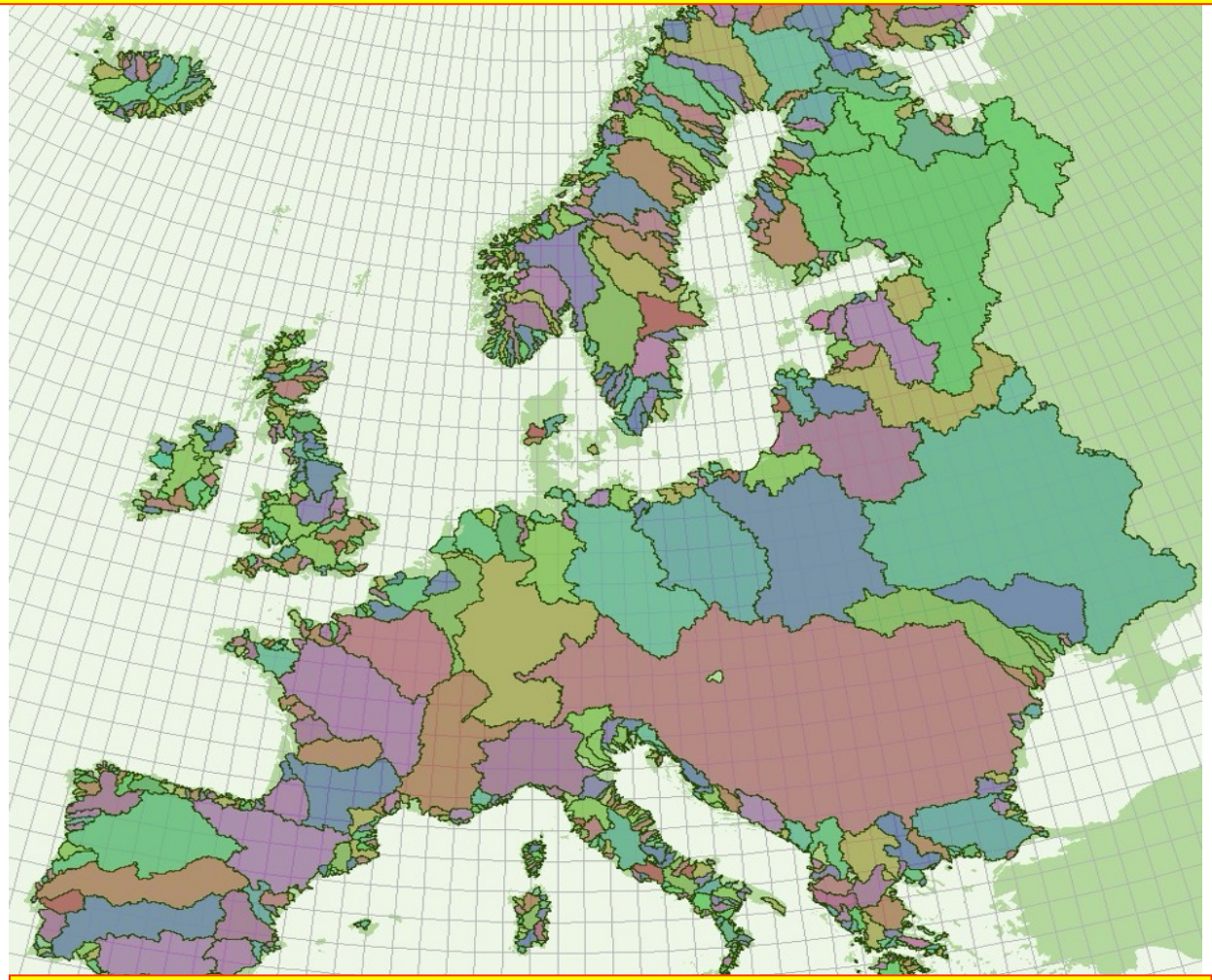
Environmental Monitoring and Management of EU Waters.

The Water Framework Directive 2000/60EC.

Integrated impact-based river basin management based on holistic assessment of surface water status.



European Rivers form a complex network of water bodies in direct contact with the coastal zones..



Ambitious objective:
“good surface water status” by 2015.

The Water Framework Directive 2000/60EC:

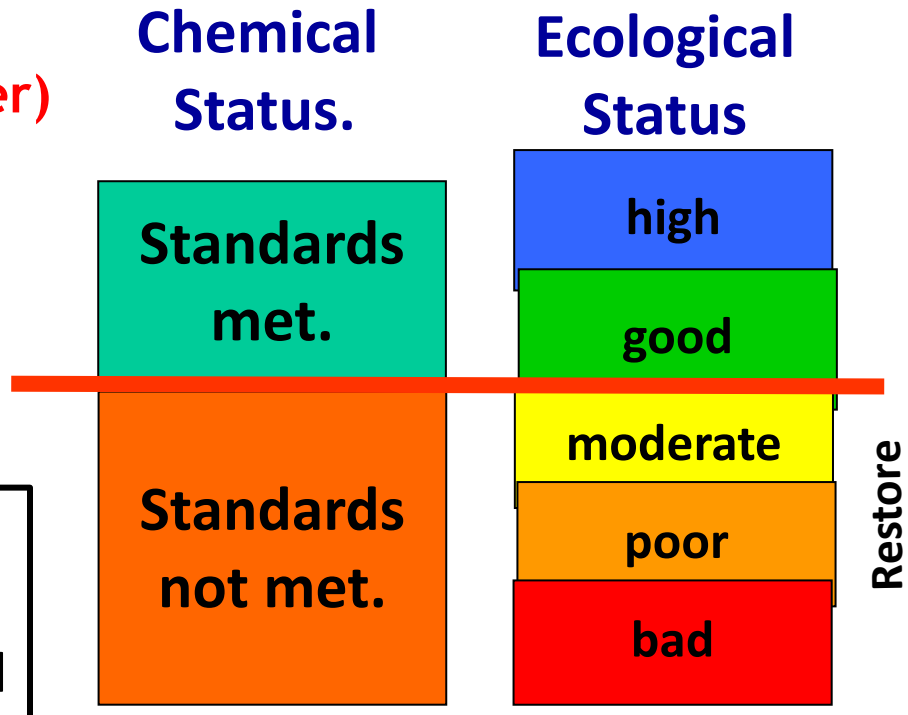
Holistic Assessment of Surface Water Status.

Chemical status (surface and ground water)

Levels of chemicals meet environmental quality standards in good status.

32.000 monitoring sites.

More than 100 chemical substances with potential harmful effects monitored regularly, e.g. nitrates, phophates, pesticides, solvents, PAHs, heavy metals.
Millions of environmental data collected every year.

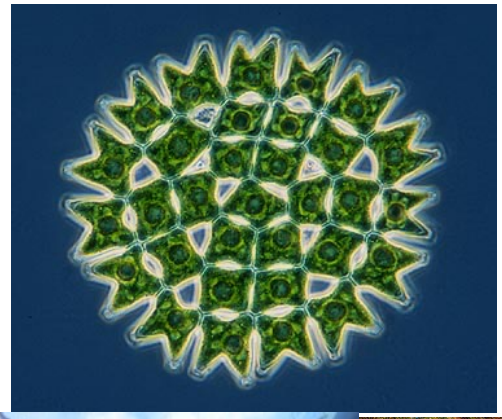


Ecological Status (surface water)

Based on biological quality indicators and ecological boundary setting.

52.000 monitoring sites.

Restore

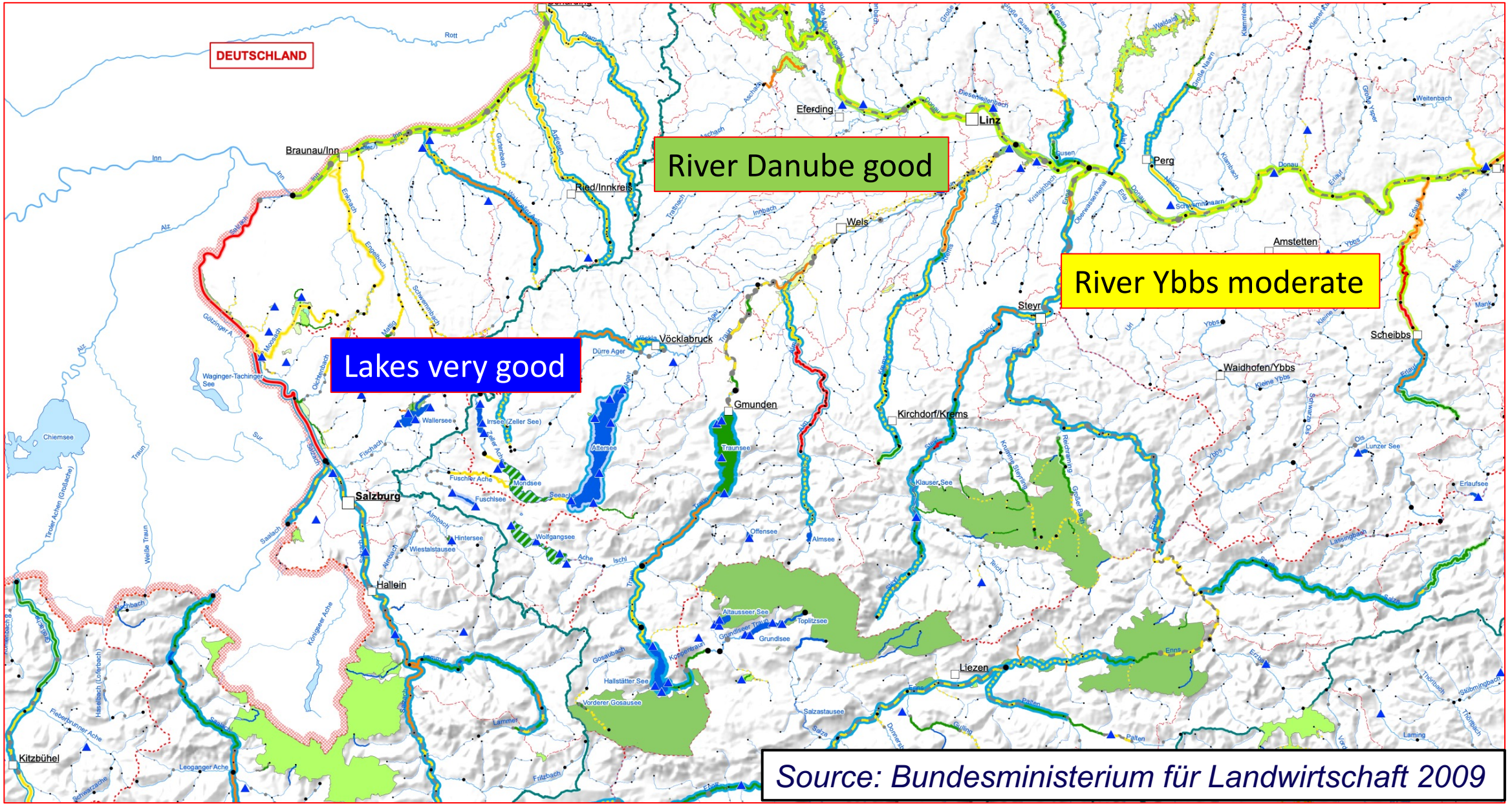


Specific indicators for each of the 7 ecoregions in Europe.

Ecosystem approach to water quality management.

Ecological Status Assessment Austria:

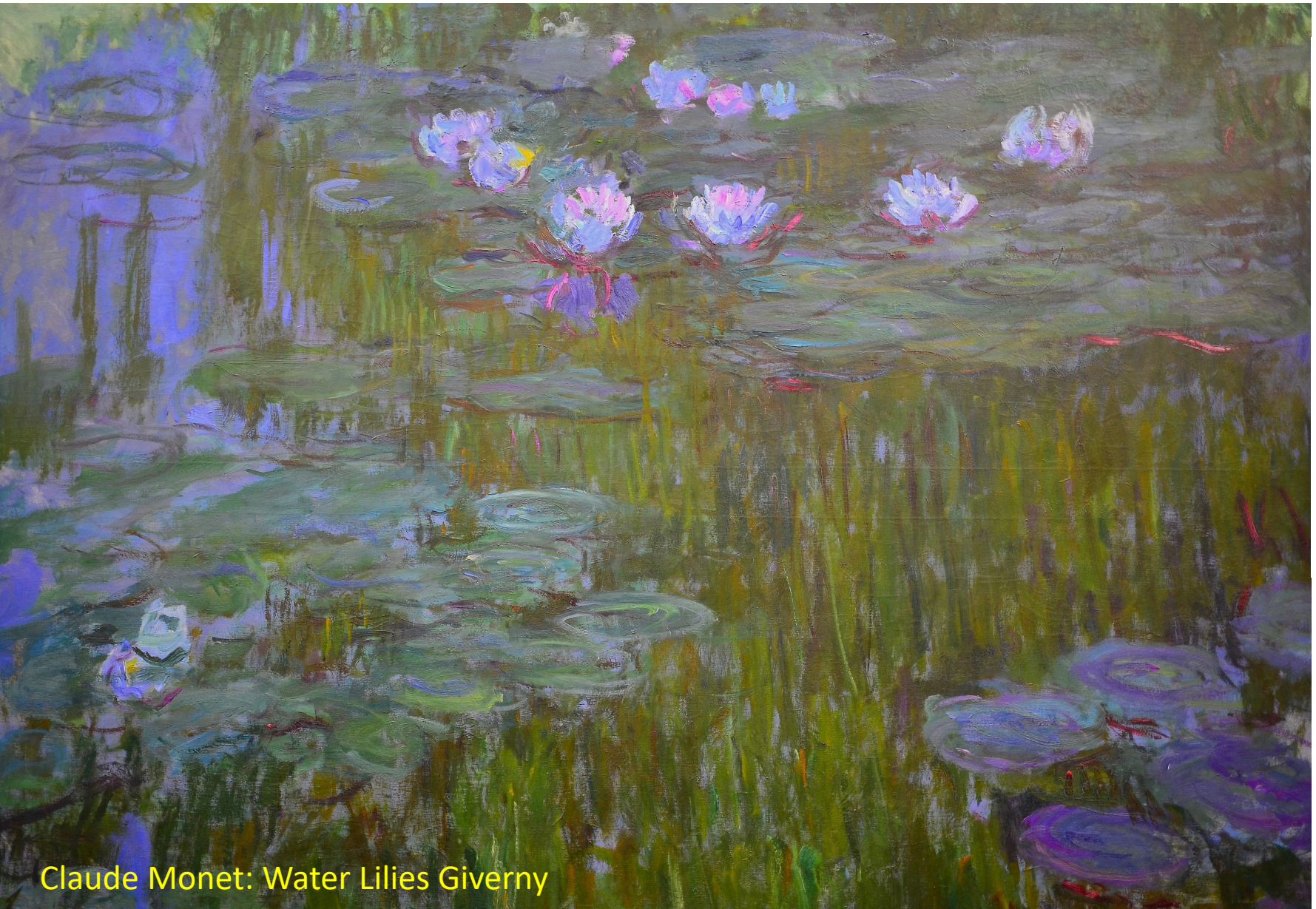
Danube, Salzkammergut Region.



Source: Bundesministerium für Landwirtschaft 2009

The Water Framework Directive 2000/60EC:

Holistic Assessment of Surface Water Status.



Claude Monet: Water Lilies Giverny

- **Status of EU <u>Surface</u> Waters 2021:**
  - 37% in good ecological status
  - 80% in good chemical status

- **Status of EU <u>Ground</u> Waters 2021:**
  - 90% in good ecological status
  - 77% in good chemical status (problem nitrates, pesticides)

Ground waters provide 70 % of the drinking water supply.

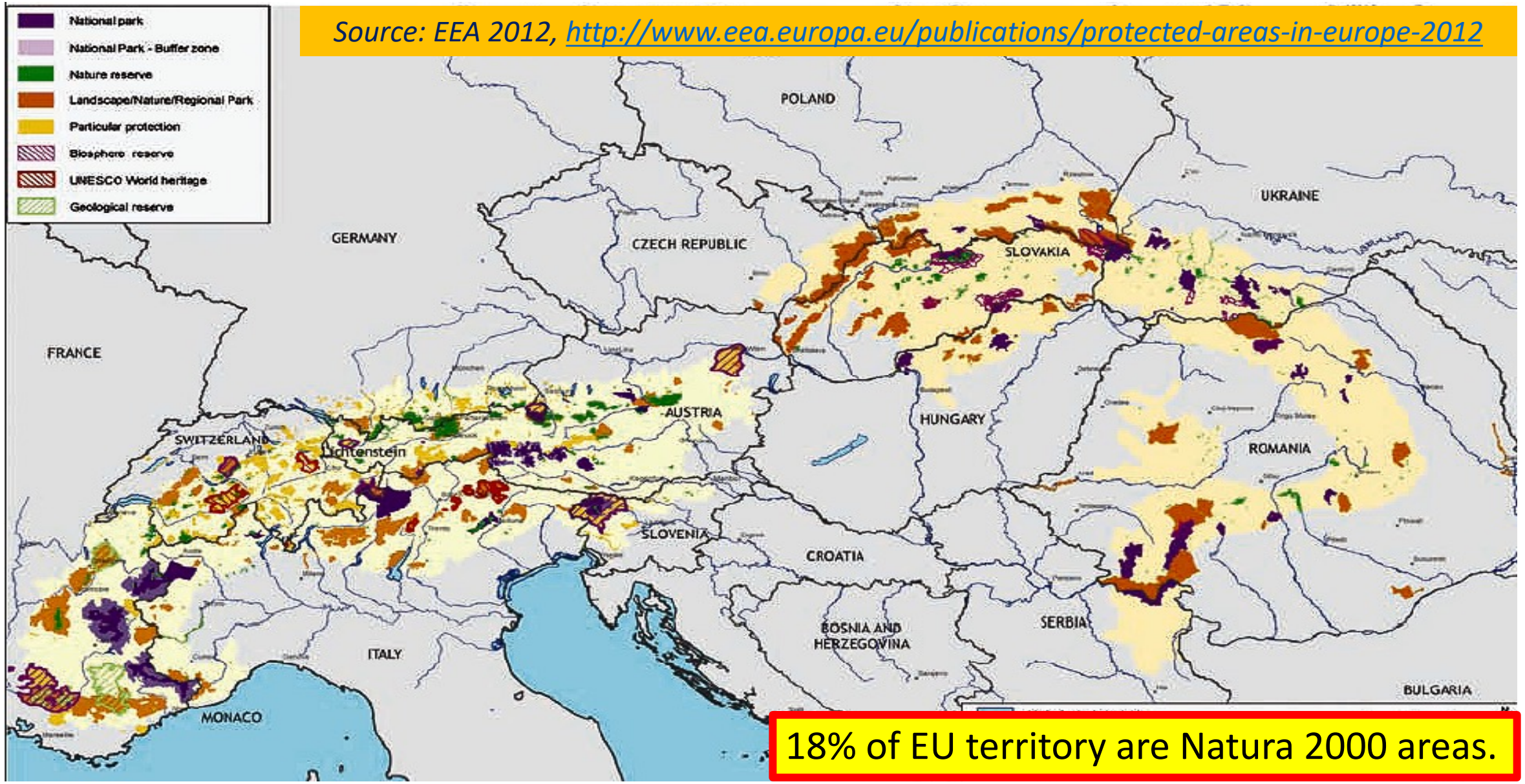
<u>Source:</u> EEA 2024

Biodiversity: EU Strategy.

- **EU Biodiversity Stategy (1998, updated 2010 and 2022):**
  - To halt the loss of biodiversity by better protection and restoration of ecosystems and sustainable agriculture, forestry and fisheries.
- **Natura 2000 areas:**
  - Conservation of natural habitats and of wild fauna and flora generating Special Areas of Conservation (SAC)

March-Thaya-Wetlands of 150 km^2 are habitat for many protected species.

Natura 2000 Areas in Alpine and Carpathian Mountains.



EU Biodiversity Strategy for 2030/2040: Renaturalisation of European Land.

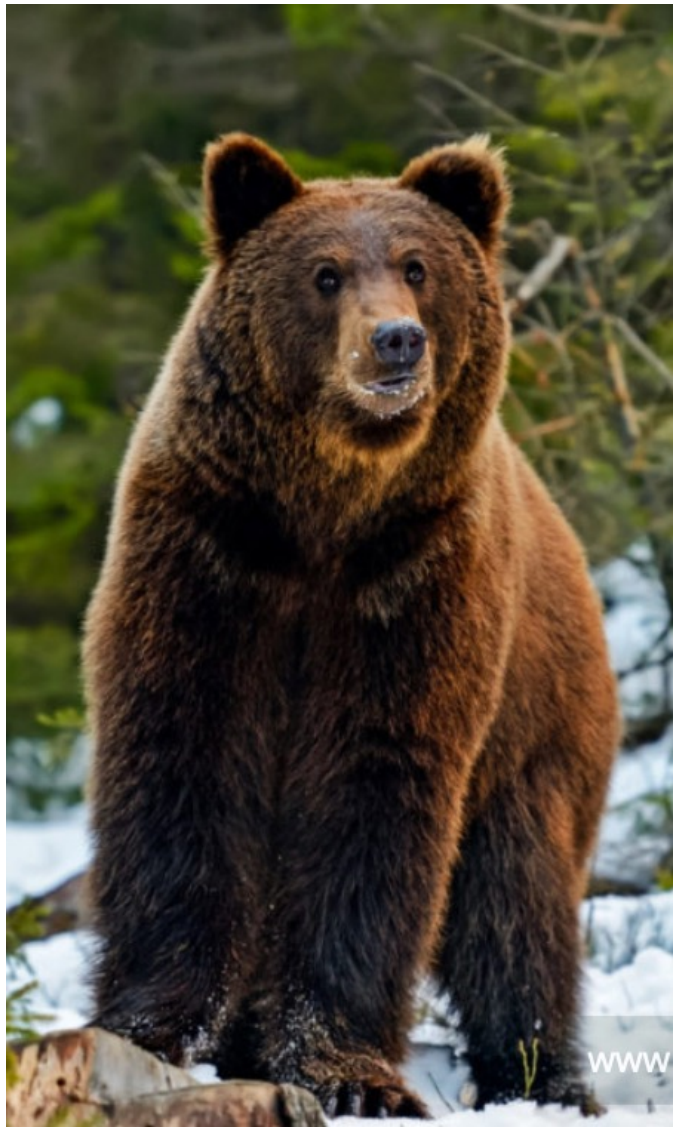
EU Nature Restoration Law for Improvement of Degraded Ecosystems (2024):

- **Goal:** increase biodiversity, reduce risk for natural disasters and achieve the climate goals.
- Binding restoration targets for specific habitats and species.
- Restoration measures should cover at least 20% of the EU's land and sea areas by 2030, and ultimately all ecosystems in need of restoration by 2050.
- Measures should aim at Green Cities, agriculture with high biodiversity features, increase of number of pollinators……defined in national implementation plans.

Specific targets:

- 30% of land and coastal areas should be protected.
- Plant 3 billion trees.
- 25.000 km free flowing rivers.
- Reduce pesticide use by 50%.
- Reduction of fertilizer use by 20%.
- 25% of agricultural area for ecological (bio-food) production.

Biodiversity:
Strict Protection of Predators?



17.000 brown bears in EU.

Difficult coexistence: 120 wolves in Austria killed 1.000 sheep (2025)

Global Warming (Climate Change).

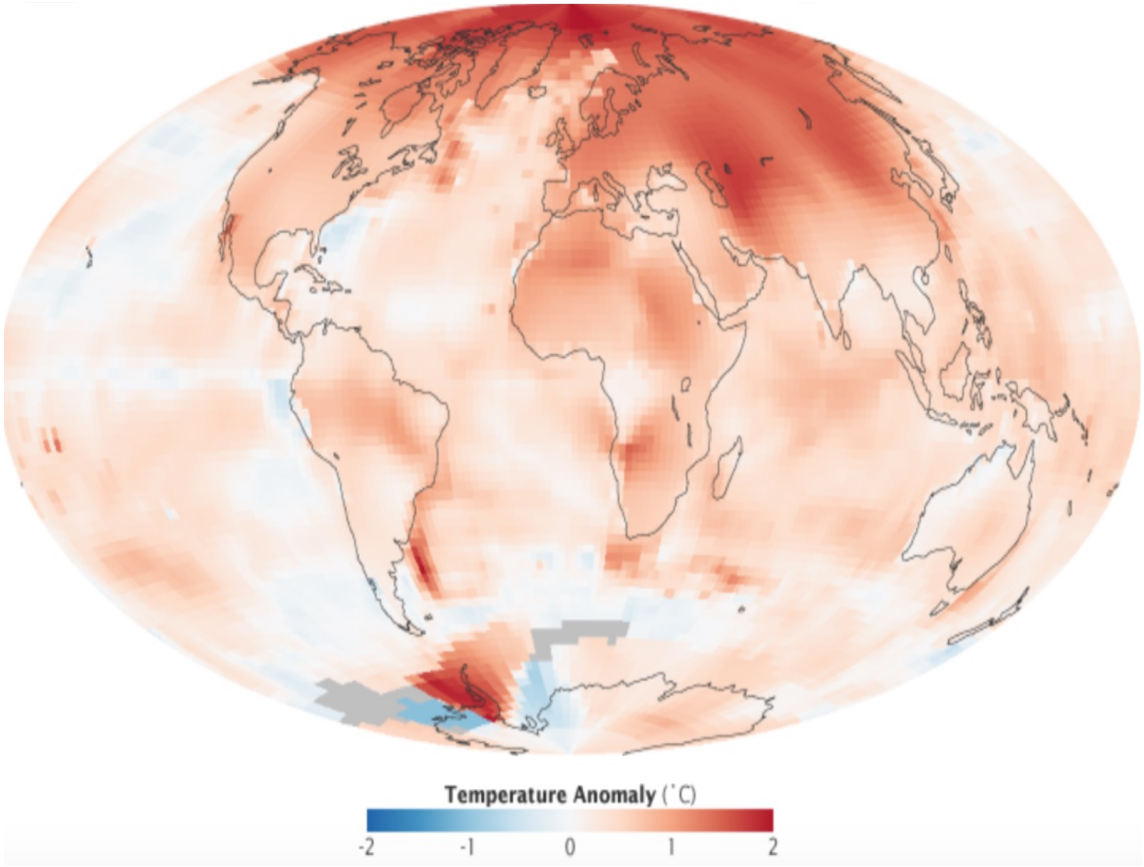
- **Evolution since preindustial times (1850):**
  - The concentration of the Green House Gases (GHGs) in the atmosphere - mainly CO_2, CH_4 and N_2O - increased from 280 to 515 ppmv CO_2eq (2024).
  - According to the IPCC this caused caused a temperature rise of 1,2°C (compared to 1850).
- **Realistic warming scenario of IPCC from 2026:**
  - By 2100 global warming could reach 2,5 – 3,0°C.
- **Combating Climate Change:**
  - Climate Agenda discussed at UN-level in Conferences of the Parties (COP1 Berlin 1997 - COP30 Berlin 2025).
  - COP21 (Paris 2015): Global agreement to set a goal of limiting global warming to **"well below 2°C"** (aiming at **1,5°C**) compared to 1850.
  - COP24 (Katowice 2018): **Limit global warming to 1,5°C.**

Global emissions of the Green House Gases must be reduced from 53 Gto/year now to 5 Gto/year in 2050 (90%).

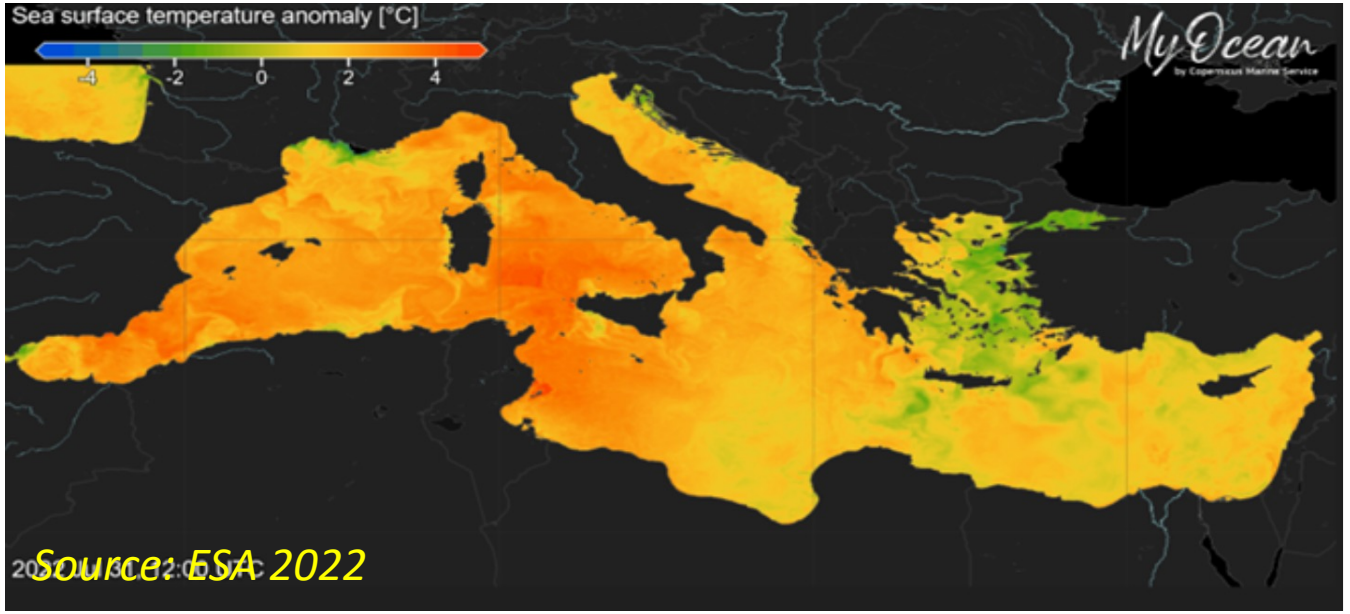
Source: IPCC and JRC

Impact of Global Warming on the Environment.

- **Basic issue for Northern hemisphere:**
  - Terrestrial warming twice as high as global average plus „Arctic Amplification.“
  - Average temperature increase in Austria 2024 = 3,6 °C (compared to 1850).



- **Increased Temperature of European Water Bodies:**
  - Mediterranean Sea temperature July 2022 up to +4,6°C.

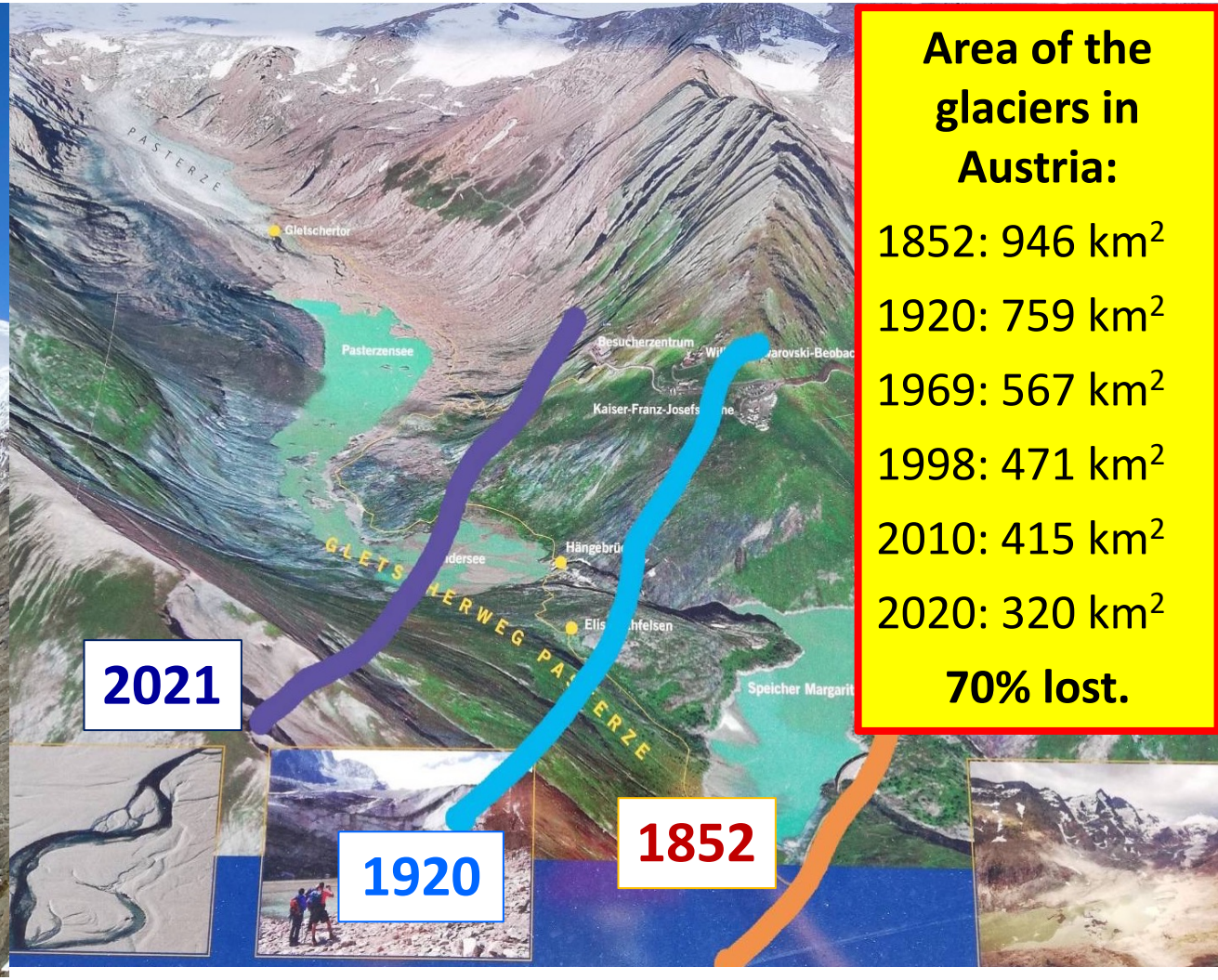


- **Possible consequences:**
  - Increased evaporation of Mediterranean and Black Sea leading to higher water content in the air and enhanced precipitation (<u>contributing</u> factor to floods).
  - Higher temperatures of lakes and rivers affects biodiversity.
  - Cooling problems for nuclear power plants.

Impact of Global Warming on the Environment.

- **Strong retreat of Alpine glaciers:**

– Glacier of Großglockner in Austrian Alps already massively reduced.



Area of the glaciers in Austria:

1852: 946 km^2

1920: 759 km^2

1969: 567 km^2

1998: 471 km^2

2010: 415 km^2

2020: 320 km^2

70% lost.

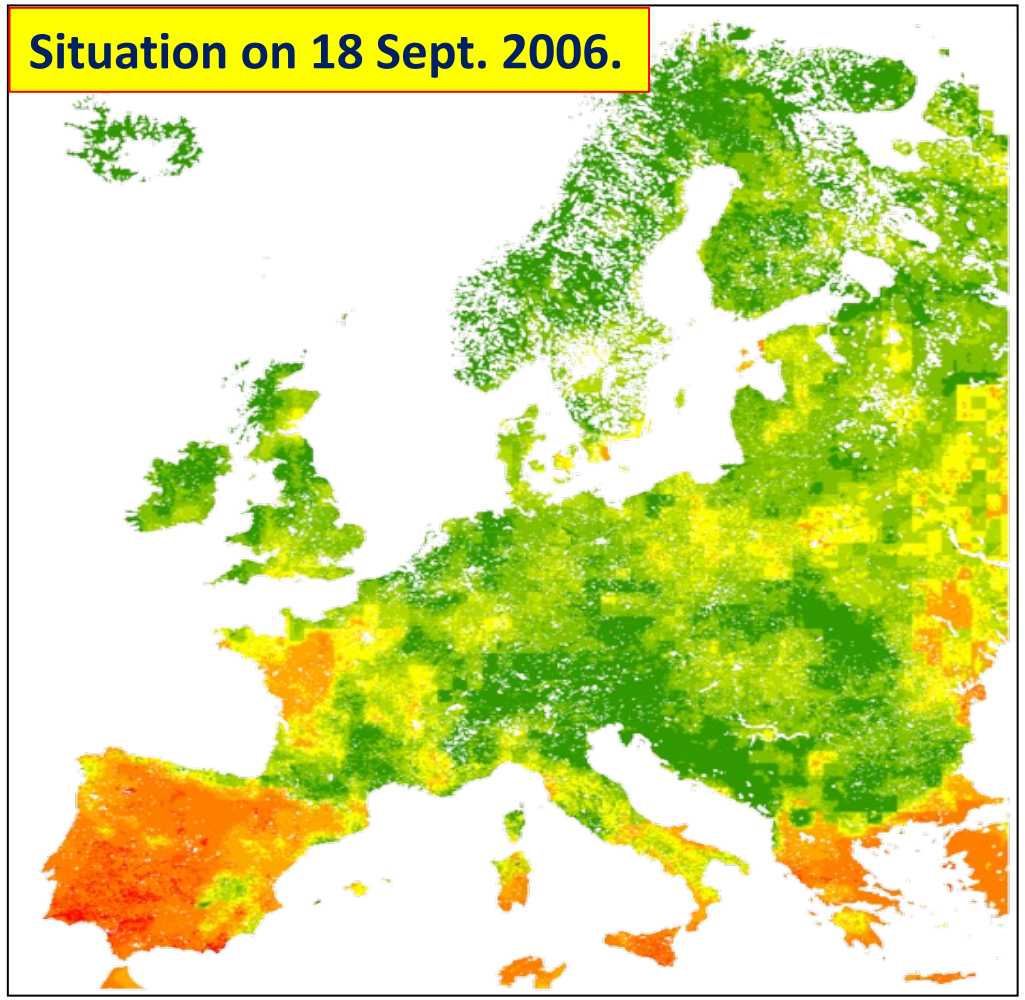
Source: Nationalpark Hohe Tauern 2022

Impact of Global Warming on the Environment.

- **Reduction in soil moisture:**
  - Reduction of agricultural yields.
  - Damages to forests.

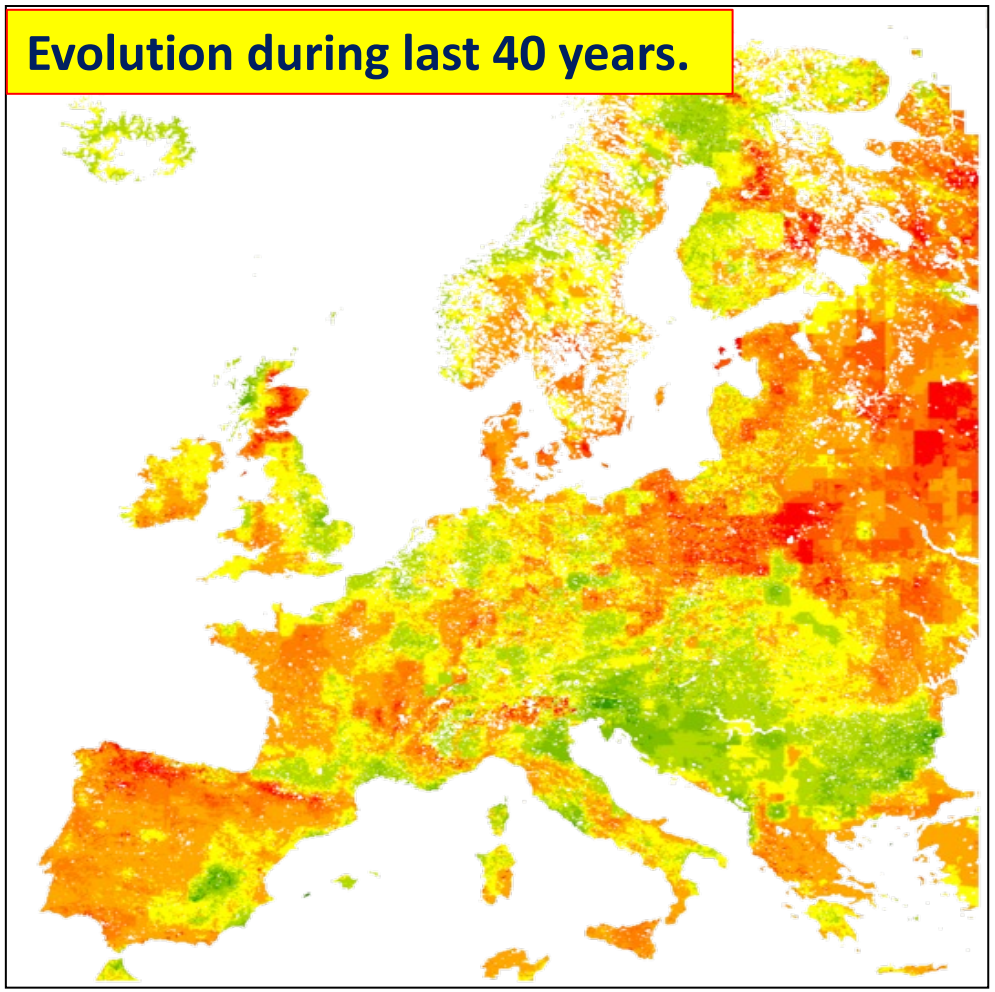
European Soil Moisture Maps.

Situation on 18 Sept. 2006.



wet normal dry

Evolution during last 40 years.



- **European soils have lost much soil humidity.**
  - Reduction of plant growth.
  - Damage to soil biodiversity.
- **Critical areas:**
  - Mediterranean land
  - Pannonian region (Weinviertel)
  - Eastern Europe

Source: JRC-IES/ ECMWF

Impact of Global Warming on the Environment.

- **Decay of Alpine forests:**

– Spreading of damaging insects like birch beetle (Borkenkäfer) killing spruce trees (Fichten).

Mölltal, Austria (Southern slope of the Alps).

EU Climate Policy.

Principles established 1990:

Reducing climate change risk

by **MITIGATION** **+** **ADAPTATION**

at the lowest cost & greatest benefit.

reducing & avoiding GHG emissions

reducing exposure & vulnerability

European Council 2019:
Decision on Green Deal „A Carbon Neutral Europe by 2050“.

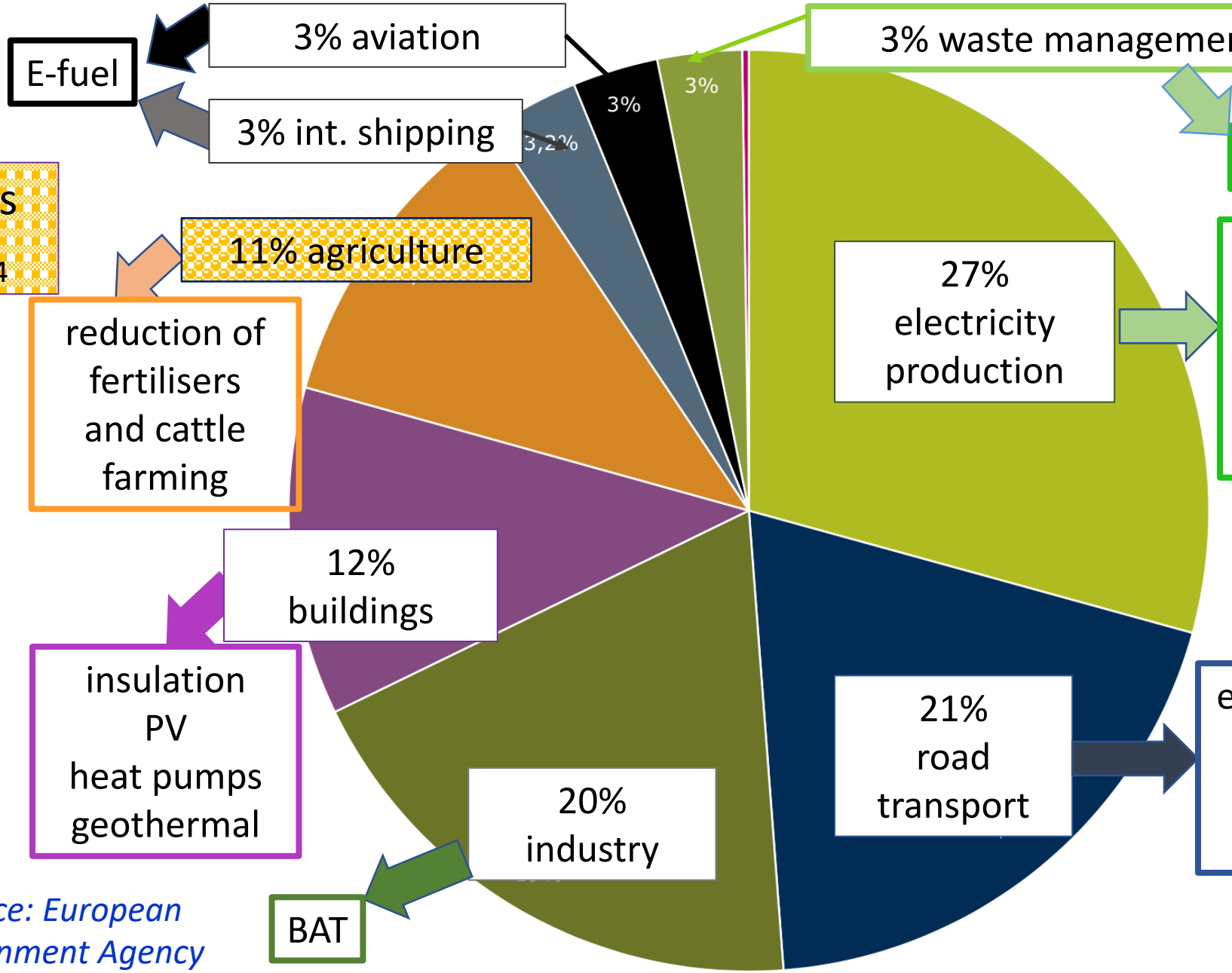
Legal targets for emission reduction of GHGs (base year 1990):

2020: minus 20% (achieved)

2030: minus 55% (measures in implementation, - 37% achieved in 2023)

2050: minus 90% (roadmap in development)

EU GHG Emissions by Sector and Transformations.



Total GHG-emissions EU27 in 2024:

3,5 Gto CO_2equ

(6% of global emissions)

Source: European Environment Agency

Towards a Climate Neutral Europe till 2050:
Implementation Tools.

A) Emissions Trading System (ETS):

- The European Commission sets maximum allowable CO_2 emissions (tons/year) for 11.000 large scale installations and aviation emitting together 45% of the EU GHGs.
- Unused emission rights can be sold, for exceedance emission certificates must be purchased.
- Should stimulate the implementation of low-carbon technologies.

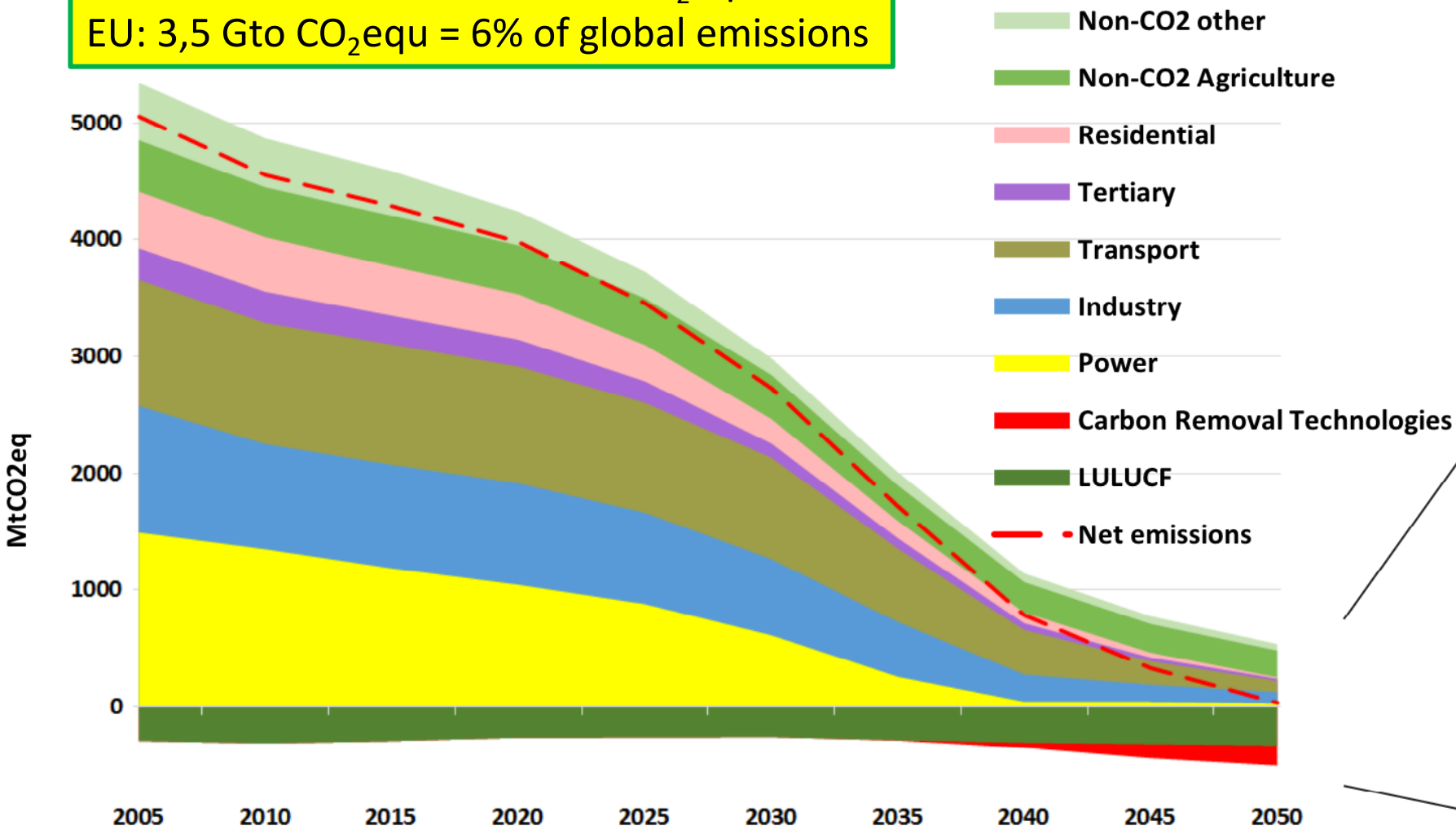
B) Effort Sharing Decision:

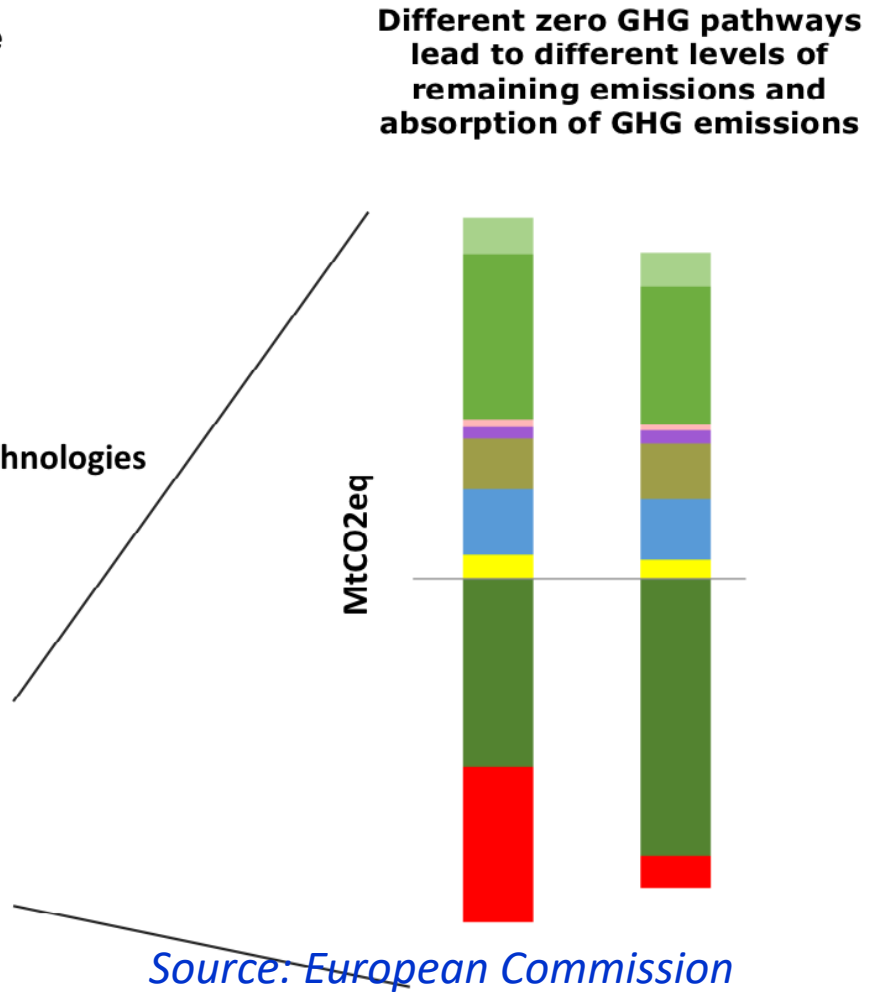
- Covers transport, buildings, agriculture, non-ETS industry (small entities) and waste, accounting for 55% of total EU emissions.
- The Effort Sharing Regulation foresees binding annual GHG emission targets for each Member State for period 2021–2030.
- Austria has a reduction goal of 50% (reference year 2005).
- **National reduction path must be established.**

Climate Policy: The EU Green Deal.

Towards a Climate Neutral Europe by 2050
EU GHG Emissions Reduction Scenario (1,5°C Goal).

Global GHG emissions: 53 Gto CO_2equ
EU: 3,5 Gto CO_2equ = 6% of global emissions





Source: European Commission

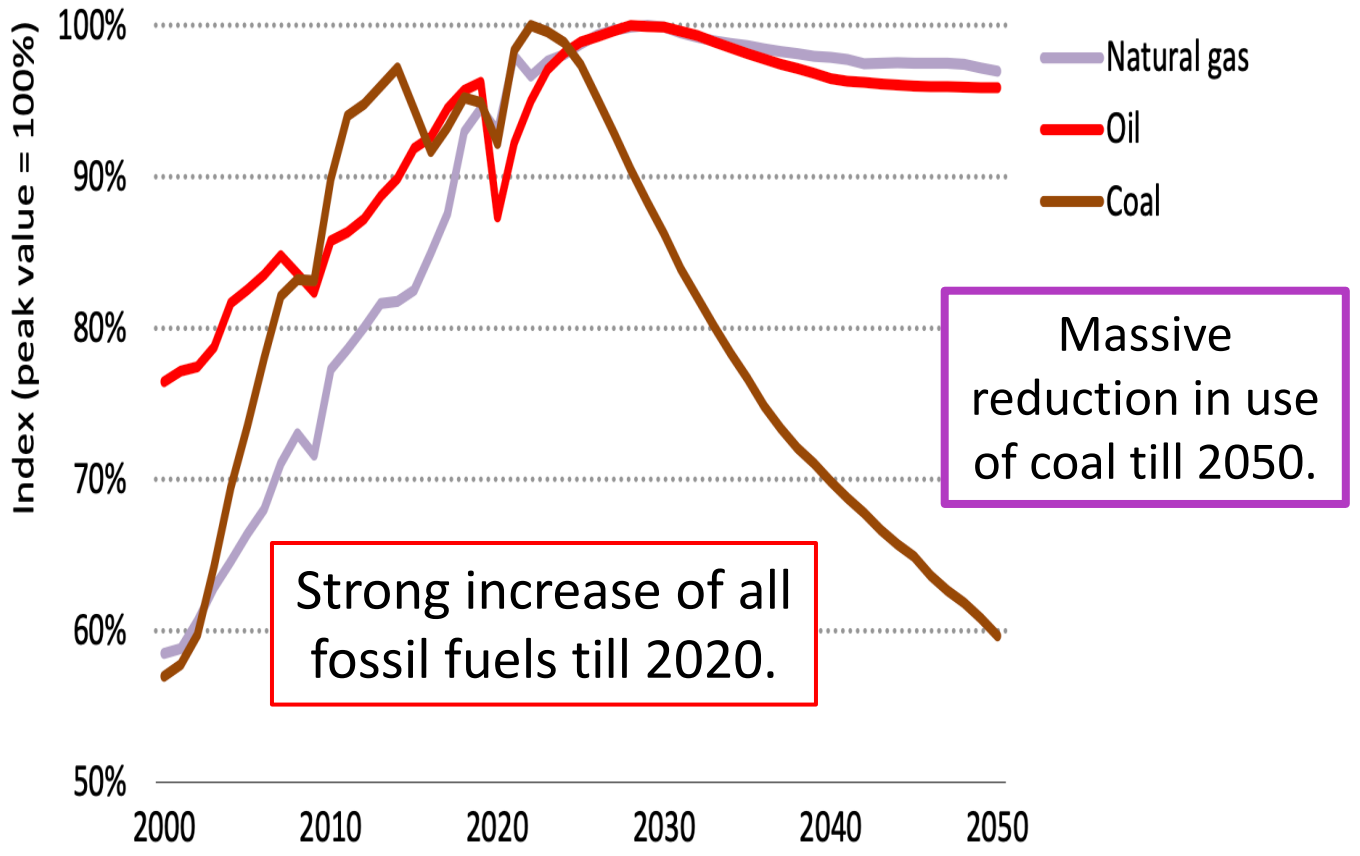
Implementation of Paris/Katowice Climate Agreement.

Can we reach the goal of limiting global warming to 1,5°?

- **Evolution of mankind till 2050:**
  - Global population will grow by 2 billion (25%).
  - Per capita economic output will increase by 85%.
  - Massive increase in energy consumption in Asia and Africa.

- **Conclusions:**
  - **The expected CO_2 emissions in 2050 will still be 25 Gto/a and not 5 Gto/a as required to achieve the 1,5 degree warming limit.**
  - **Average global warming expected to be +3°C.**

Constant consumption of gas and oil till 2050.



Source: International Energy Agency 2024

IEA. CC BY 4.0.

Sustainable Development.

Source: United Nations

- On a global level the European Union is actively pursuing the achievement of the UN Sustainable Development Goals established 2015.
- Companies have to report on their sustainability performance.

Social Policies.

Social standards are not a community matter.

- Member states have so far refused to develop common social standards.
- Has proven to be impossible due to a huge difference in the social and economic situation in the Member States.
- A few common achievements have been made however:

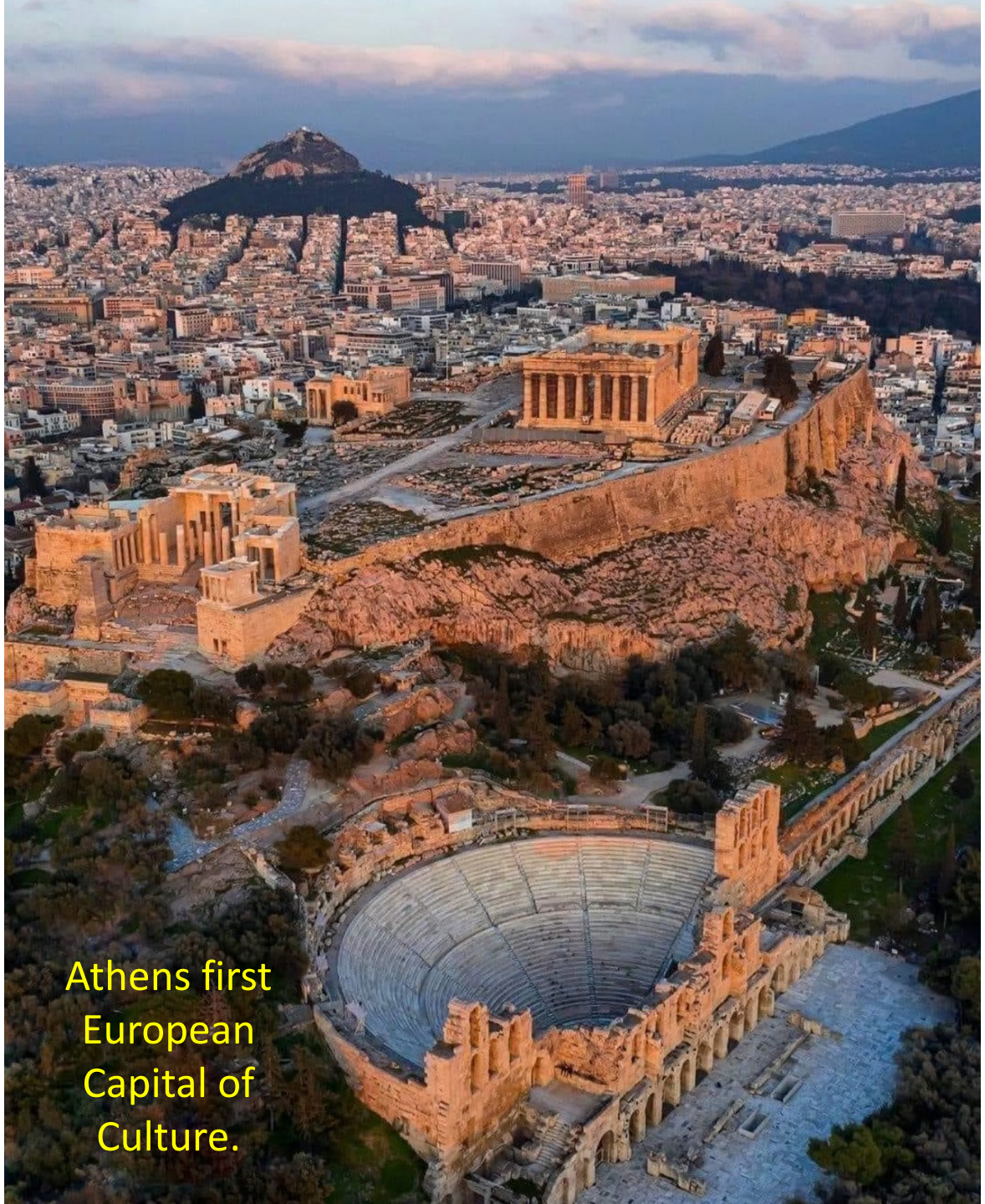
– Social security coordination of pension rights acquired in different EU member states.

– Equal rights for both fathers and mothers to time off work following the arrival of a baby.

– The new Commission has however promised to make a drive for more social cohesion, e.g. by the introduction of EU minimum wages (adapted to the actual costs of living in that Member State).

- Health issues are part of social policies – no common standards or policies.
- Proved to be a huge problem in the present Corona pandemic – lack of coordinated action of EU Member States.

The large differences in social standards are a major driving force for the movement of people within the EU.



Social Policies.

The European Capital of Culture program was launched in the summer of 1985.

- **The European Capitals of Culture (ECOC) initiative is designed to:**
  - Highlight the richness and diversity of cultures in Europe.
  - Celebrate the cultural features Europeans share.
  - Increase European citizens' sense of belonging to a common cultural area.
  - Foster the contribution of culture to the development of cities.
  - Austria: Graz, Linz, Bad Ischl.